

KLEINE BROGEL AB, Belgium

WMO: 064790

Lat: 51.1689N

Lon: 5.4628E

Elev: 61

StdP: 100.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-5.0	-10.5	1.5	-5.2	-8.0	1.9	-3.5	11.8	9.7	10.6	9.7	1.8	70	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	30.9	20.2	28.6	19.1	26.5	18.4	21.3	28.0	20.3	26.6	19.3	24.8	2.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.0	23.5	18.1	13.2	22.3	17.3	12.4	21.4	61.9	28.0	58.3	26.8	55.2	24.9	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.0	6.9	-11.0	35.2	3.5	2.1	-13.5	36.7	-15.5	38.0	-17.4	39.2	-19.9	40.7		
(5)			-11.3	23.5	3.4	1.2	-13.7	24.3	-15.7	25.0	-17.6	25.7	-20.0	26.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	3.2	4.0	6.7	10.2	13.8	16.7	18.8	18.3	15.1	11.3	6.9	3.9	(6)	
	DBStd	6.61	4.35	4.05	3.60	3.76	3.50	3.34	3.30	2.94	2.87	3.49	3.69	4.17	(7)	
	HDD10.0	873	212	171	113	44	7	0	0	0	1	27	106	193	(8)	
	HDD18.3	2884	469	402	362	246	147	71	34	37	104	218	344	448	(9)	
	CDD10.0	1151	1	2	10	49	123	202	274	256	153	67	12	3	(10)	
	CDD18.3	119	0	0	0	1	5	22	49	34	7	1	0	0	(11)	
	CDH23.3	1367	0	0	0	28	101	267	518	373	76	4	0	0	(12)	
	CDH26.7	436	0	0	0	4	21	78	184	134	16	0	0	0	(13)	
(14) Wind	WSAvg	2.9	3.5	3.4	3.3	2.8	2.8	2.7	2.6	2.4	2.5	2.8	2.9	3.4	(14)	
(15) Precipitation	PrecAvg	817	69	67	54	47	59	77	79	80	62	67	73	83	(15)	
	PrecMax	1055	143	156	117	99	116	225	169	179	175	185	130	156	(16)	
	PrecMin	632	1	11	0	3	20	14	21	21	16	25	12	28	(17)	
	PrecStd	102	33	33	28	22	25	41	38	46	40	30	29	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	15.0	20.3	26.8	29.3	32.3	34.6	33.6	28.8	23.4	17.3	13.8	(19)	
		MCWB	10.8	9.7	12.1	15.4	18.1	20.3	20.9	21.0	19.1	17.3	13.8	11.5	(20)	
	2%	DB	11.6	12.4	17.1	23.2	25.9	28.8	31.0	30.4	25.5	20.4	15.0	12.4	(21)	
		MCWB	9.9	9.3	11.0	14.1	16.4	19.1	20.4	20.2	18.2	16.1	12.6	10.8	(22)	
	5%	DB	10.2	11.0	14.4	20.2	23.6	26.4	28.5	27.6	22.7	18.2	13.6	11.2	(23)	
		MCWB	8.7	8.7	10.0	12.8	15.5	18.2	19.3	19.4	17.1	15.1	11.7	9.7	(24)	
	10%	DB	8.9	9.7	12.4	17.4	21.2	24.1	26.2	24.8	20.5	16.5	12.2	9.7	(25)	
		MCWB	7.6	8.0	9.2	11.5	14.6	17.1	18.7	18.2	15.8	13.9	10.7	8.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	11.0	13.2	16.3	19.8	22.0	22.7	22.7	20.5	17.9	14.4	12.2	(27)	
		MCDB	12.5	13.0	17.9	24.0	26.5	29.6	30.6	30.0	26.1	22.2	16.3	13.1	(28)	
	2%	WB	10.1	10.1	11.9	14.7	17.7	20.3	21.4	21.0	19.0	16.6	13.0	11.1	(29)	
		MCDB	11.2	11.7	15.3	21.1	23.6	26.6	28.3	27.5	23.7	19.6	14.7	12.3	(30)	
	5%	WB	8.9	9.1	10.7	13.3	16.3	19.1	20.3	19.9	17.8	15.4	11.9	9.9	(31)	
		MCDB	10.1	10.6	13.6	18.9	21.5	24.6	26.8	26.3	21.8	17.8	13.3	11.0	(32)	
	10%	WB	7.7	8.0	9.6	12.1	15.2	17.9	19.3	18.9	16.6	14.4	10.7	8.6	(33)	
		MCDB	8.8	9.7	12.2	16.5	20.0	22.8	24.6	23.8	19.7	16.3	12.1	9.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.5	8.8	11.3	11.1	11.4	11.1	11.1	10.0	8.3	6.1	4.9	(35)	
		MCDBR	5.7	7.9	12.9	16.1	15.8	15.5	15.6	15.9	13.5	10.4	7.1	5.3	(36)	
	5% WB	MCWBR	4.8	5.5	7.6	8.2	7.4	7.1	6.2	6.8	6.9	6.4	5.2	4.4	(37)	
		MCWBR	5.5	6.5	10.6	14.1	13.2	13.4	13.6	13.7	11.7	9.4	6.5	5.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.329	0.353	0.385	0.417	0.424	0.422	0.429	0.419	0.391	0.367	0.342	0.326	(40)	
		taud	2.380	2.341	2.255	2.199	2.222	2.244	2.228	2.277	2.345	2.398	2.384	2.349	(41)	
	Ebn at Noon		683	764	803	820	832	836	822	809	791	739	657	617	(42)	
		Edh at Noon	63	85	110	130	133	132	132	119	100	79	63	56	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.76	1.44	2.60	4.15	4.97	5.39	5.08	4.33	3.22	1.90	0.89	0.59	(44)	
	RadStd		0.10	0.28	0.37	0.46	0.44	0.45	0.61	0.41	0.31	0.15	0.10	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (89 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page